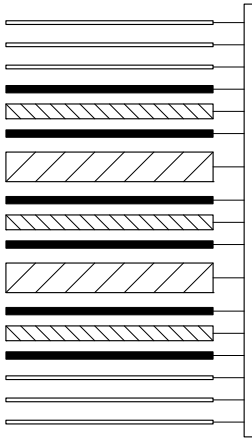
BOARD NAME FABRICATION DOCUMENT

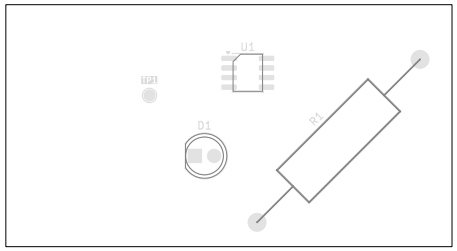
LAYER STACKUP LEGEND



Material	Layer	Thickness	Dielectric	Type	Gerber
	F.Paste			Paste Mask	
	F.Silkscreen		Direct Printing	Legend	GBR
	F.Mask	0.02mm	Solder Resist	Solder Mask	GBR
Copper	**L1 (Sig, PWR)**	**0.07mm (2.00oz)**		**Signal**	**GBR**
Prepreg		0.18mm	FR4_7628	Dielectric	
Copper	**L2 (GND)**	**0.035mm (1.00oz)**		**Plane**	**GBR**
Core		0.4mm	FR4	Dielectric	
Copper	**L3 (Sig, PWR)**	**0.035mm (1.00oz)**		**Signal**	**GBR**
Prepreg		0.18mm	FR4_7628	Dielectric	
Copper	**L4 (Sig, PWR)**	**0.035mm (1.00oz)**		**Signal**	**GBR**
Core		0.4mm	FR4	Dielectric	
Copper	**L5 (GND)**	**0.035mm (1.00oz)**		**Plane**	**GBR**
Prepreg		0.18mm	FR4_7628	Dielectric	
Copper	**L6 (Sig, PWR)**	**0.07mm (2.00oz)**		**Signal**	**GBR**
	B.Mask	0.02mm	Solder Resist	Solder Mask	GBR
	B.Silkscreen		Direct Printing	Legend	GBR
	B.Paste			Paste Mask	

Total thickness: 1.66mm
Note: external layer thicknesses are specified after plating

TOP FABRICATION (SCALE 1:1)



IMPEDANCE TABLE

Transmission Line	Impedance [ohms]	Tolerance [ohms]	Layer	Trace Width [mm]	Gap [mm]	Ref. Layers
Edge-Coupled Coated Microstrip	100	±10 %	L1	0.2032	0.28	L2

FABRICATION NOTES (UNLESS OTHERWISE SPECIFIED)

1) FABRICATE PER IPC-6012A CLASS 2.

2) OUTLINE DEFINED IN SEPARATE GERBER FILE WITH "Edge_Cuts.GBR" SUFFIX.

   DIMENSIONS OF CIRCUMSIZED RECTANGLE SHOWN ON THIS DRAWING FOR REFERENCE ONLY.

3) SEE SEPARATE DRILL FILES WITH ".DRL" SUFFIX FOR HOLE LOCATIONS.

   SELECTED HOLE LOCATIONS SHOWN ON THIS DRAWING FOR REFERENCE ONLY.

4) SURFACE FINISH: IMMERSION GOLD

5) SOLDERMASK ON BOTH SIDES OF THE BOARD SHALL BE LPI, COLOR BLACK.

6) SILK SCREEN LEGEND TO BE APPLIED PER LAYER STACKUP USING YELLOW NON-CONDUCTIVE EPOXY INK.

7) ALL VIAS ARE TENTED ON BOTH SIDES UNLESS SOLDERMASK OPENED IN GERBER.

8) VENDOR SHOULD FOLLOW ROHS COMPLIANT PROCESS AND Pb FREE FOR MANUFACTURING

9) PCB MATERIAL REQUIREMENTS:

   A. FLAMMABILITY RATING MUST MEET OR EXCEED UL94V-0 REQUIREMENTS.
   B. Tg 170 C OR EQUIVALENT.
   C. EQUIVALENT MATERIAL SHALL BE RoHS COMPLIANT, HALOGEN FREE AND APPROVED BY COMPANY.

10) DESIGN GEOMETRY MINIMUM FEATURE SIZES:

BOARD SIZE	59.000 × 32.000 mm
BOARD THICKNESS	1.660 mm
TRACE WIDTH	N/A mm
TRACE TO TRACE	0.200 mm
MIN. HOLE (PTH)	0.900 mm
MIN. HOLE (NPTH)	N/A mm
ANNULAR RING	0.450 mm
COPPER TO HOLE	0.254 mm
COPPER TO EDGE	0.250 mm
HOLE TO HOLE	0.254 mm

11) REFER TO IMPEDANCE TABLE FOR IMPEDANCE CONTROL REQUIREMENT

12) CONFIRM SPACE WIDTHS AND SPACINGS.

ALL DIMENSIONS IN MILLIMETERS UNLESS OTHERWISE SPECIFIED.

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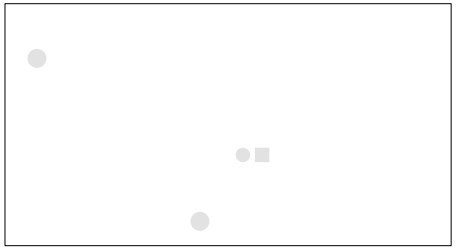
DRN 2024-04-13 F. LAST	THIS DESIGN AND/OR DRAWING IS THE PROPERTY OF RYAN DYNAMICS AND SHALL NOTBE REPRODUCED WITHOUT AUTHORISATION.	
CHK		
ENG APP	BOARD NAME	
MFR APP		
SHEET PATH	GIT HASH 0e558a0	DRAWING No FABRICATION
FILENAME boardwright.kicad_pcb	VARIANT CHECKED	REVISION 0.1.1 SHEET 1 OF 11

A4

BOARD NAME FABRICATION DOCUMENT

BOTTOM FABRICATION (SCALE 1:1)



ALL DIMENSIONS IN MILLIMETERS UNLESS OTHERWISE SPECIFIED.

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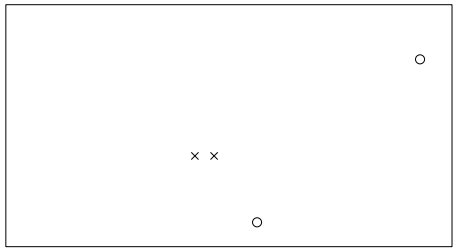
ENG APP	BOARD NAME		
MFR APP			
SHEET PATH	GIT HASH 0e558a0	DRAWING No FABRICATION	A4
FILENAME boardwright.kicad_pcb	VARIANT CHECKED	REVISION 0.1.1 / SHEET 2 OF 11	

BOARD NAME FABRICATION DOCUMENT

DRILL TABLE

Symbol	Count	Hole Size	Plated	Hole Shape	Drill Layer Pair	Hole Type
×	2	0.90mm (35.43mils)	PTH	Round	L1 (Sig, PWR) - L6 (Sig, PWR)	Pad
○	2	1.20mm (47.24mils)	PTH	Round	L1 (Slg, PWR) - L6 (Slg, PWR)	Pad
	Total 4					

DRILL DRAWING L1 – L6 (SCALE 1:1)



PROJECT PROJECT NAME	Ryan Dynamics THIS DESIGN AND/OR DRAWING IS THE PROPERTY OF RYAN DYNAMICS AND SHALL NOTBE REPRODUCED WITHOUT AUTHORISATION.		
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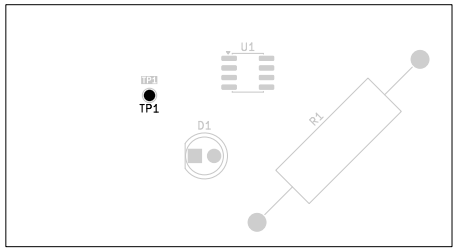
CHK			
ENG APP	BOARD NAME		
MFR APP			
SHEET PATH	GIT HASH 0e558a0	DRAWING No FABRICATION	A4
FILENAME boardwright.kicad_pcb	VARIANT CHECKED	REVISION 0.1.1 SHEET 3 OF 11	

BOARD NAME FABRICATION DOCUMENT

TOP TEST POINTS (SCALE 1:1)

Ref.	Net	X [mm]	Y [mm]
TP1	unconnected-(TP1-Pad1)	81.00	34.00

Ref.	Net	X [mm]	Y [mm]



ALL DIMENSIONS IN MILLIMETERS UNLESS OTHERWISE SPECIFIED.

PROJECT PROJECT NAME	Ryan Dynamics		
DRN 2024-04-13 F. LAST	THIS DESIGN AND/OR DRAWING IS THE PROPERTY OF RYAN DYNAMICS AND SHALL NOTBE REPRODUCED WITHOUT AUTHORISATION.		
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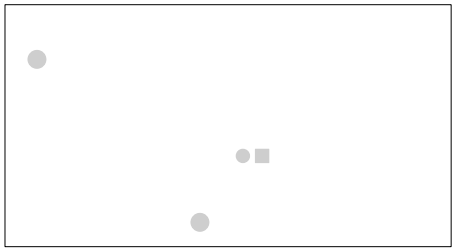
ENG APP	BOARD NAME		
MFR APP			
SHEET PATH	GIT HASH 0e558a0	DRAWING No FABRICATION	A4
FILENAME boardwright.kicad_pcb	VARIANT CHECKED	REVISION 0.1.1 / SHEET 4 OF 11	

BOARD NAME FABRICATION DOCUMENT

BOTTOM TEST POINTS (SCALE 1:1)

Ref.	Net	X [mm]	Y [mm]



ALL DIMENSIONS IN MILLIMETERS UNLESS OTHERWISE SPECIFIED.

PROJECT PROJECT NAME	Ryan Dynamics		
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SHEET PATH	GIT HASH 0e558a0	DRAWING No FABRICATION	A4
FILENAME boardwright.kicad_pcb	VARIANT CHECKED	REVISION 0.1.1 / SHEET 5 OF 11	

BOARD NAME FABRICATION DOCUMENT

L1 (Sig, PWR) (SCALE 1:1)

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CHK			
ENG APP	BOARD NAME		
MFR APP			
SHEET PATH	GIT HASH 0e558a0	DRAWING No FABRICATION	A4
FILENAME boardwright.kicad_pcb	VARIANT CHECKED	REVISION 0.1.1 / SHEET 6 OF 11	

BOARD NAME FABRICATION DOCUMENT

L2 (GND) (SCALE 1:1)

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CHK			
ENG APP	BOARD NAME		
MFR APP			
SHEET PATH	GIT HASH 0e558a0	DRAWING No FABRICATION	A4
FILENAME boardwright.kicad_pcb	VARIANT CHECKED	REVISION 0.1.1 / SHEET 7 OF 11	

BOARD NAME FABRICATION DOCUMENT

L3 (Sig, PWR) (SCALE 1:1)

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CHK			
ENG APP	BOARD NAME		
MFR APP			
SHEET PATH	GIT HASH 0e558a0	DRAWING No FABRICATION	A4
FILENAME boardwright.kicad_pcb	VARIANT CHECKED	REVISION 0.1.1 / SHEET 8 OF 11	

BOARD NAME FABRICATION DOCUMENT

L4 (Sig, PWR) (SCALE 1:1)

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DRN 2024-04-13 F. LAST			
CHK			
ENG APP	BOARD NAME		
MFR APP			
SHEET PATH	GIT HASH 0e558a0	DRAWING No FABRICATION	A4
FILENAME boardwright.kicad_pcb	VARIANT CHECKED	REVISION 0.1.1 / SHEET 9 OF 11	

BOARD NAME FABRICATION DOCUMENT

L5 (GND) (SCALE 1:1)

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CHK			
ENG APP	BOARD NAME		
MFR APP			
SHEET PATH	GIT HASH 0e558a0	DRAWING No FABRICATION	A4
FILENAME boardwright.kicad_pcb	VARIANT CHECKED	REVISION 0.1.1 SHEET 10 OF 11	

BOARD NAME FABRICATION DOCUMENT

L6 (Sig, PWR) (SCALE 1:1)

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CHK			
ENG APP	BOARD NAME		
MFR APP			
SHEET PATH	GIT HASH 0e558a0	DRAWING No FABRICATION	A4
FILENAME boardwright.kicad_pcb	VARIANT CHECKED	REVISION 0.1.1 / SHEET 11 OF 11	